Measurement of Gravitational Acceleration Using a Simple Pendulum

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Date: March 25, 2026

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Abstract

This experiment investigates the relationship between the period of a simple pendulum and its length. We measured the period for various lengths and calculated the gravitational acceleration as $g = (9.81 \pm 0.05)\,\mathrm{m/s^2}$.

1. Introduction

The purpose of this experiment is to measure the gravitational acceleration g by analyzing the period of a simple pendulum. The theoretical relationship governing the period T of a simple pendulum is given by the equation:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

From this, we can rearrange to find g:

$$g = \frac{4\pi^2 L}{T^2}$$

We hypothesize that the measured value of g will be close to the standard value of $9.81\,\mathrm{m/s^2}$.

2. Experimental Setup

The apparatus used in this experiment includes: - A rigid support to suspend the pendulum. - A mass (bob) attached to a string of varying lengths. - A stopwatch for measuring the time of oscillations.

The procedure involved measuring the time taken for 50 complete oscillations for pendulum lengths of 0.5 m, 1.0 m, 1.5 m, and 2.0 m. The lengths were measured using a meter stick, with an uncertainty of ± 0.01 m.

3. Data and Results

The following table summarizes the measured times for 50 oscillations and the corresponding periods:

Length L (m)	Time for 50 oscillations t_{50} (s)	Period T (s)
0.5	70.0	1.400
1.0	100.0	2.000
1.5	122.5	2.450
2.0	140.0	2.800

Table 1: Measured times and calculated periods for different pendulum lengths.

The mean period for each length was calculated, and the standard deviation was found to be minimal, indicating consistent measurements.

4. Error Analysis

The primary sources of error in this experiment include: - Timing inaccuracies due to human reaction time. - Variations in the length of the pendulum due to measurement error.

The propagation of uncertainty in g can be expressed as:

$$\delta g = g\sqrt{\left(\frac{\delta L}{L}\right)^2 + \left(\frac{2\delta T}{T}\right)^2}$$

Using the measured values, we find:

$$g = (9.81 \pm 0.05)\,\mathrm{m/s^2}$$

5. Discussion

The calculated value of g is consistent with the theoretical value of $9.81\,\mathrm{m/s^2}$, with a percentage error of approximately 0.51%. The dominant sources of error were identified as timing inaccuracies and slight variations in the length of the pendulum. To reduce these errors, a more precise timing method, such as using electronic sensors, could be employed.

6. Conclusion

The experiment successfully measured the gravitational acceleration to be $g = (9.81 \pm 0.05)\,\mathrm{m/s^2}$, which is in good agreement with the expected theoretical value. This confirms the reliability of the simple pendulum as a method for measuring gravitational acceleration.

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